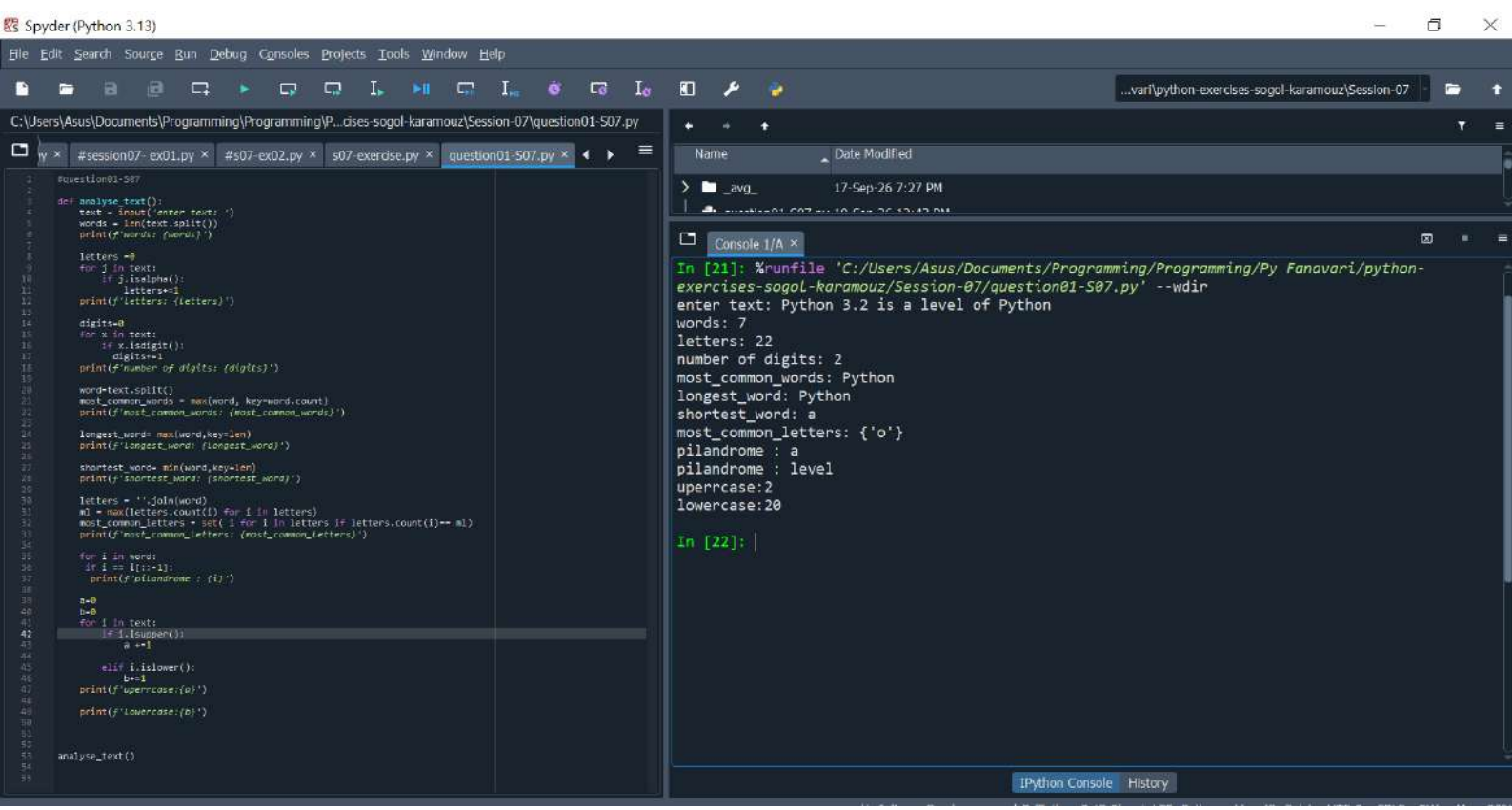
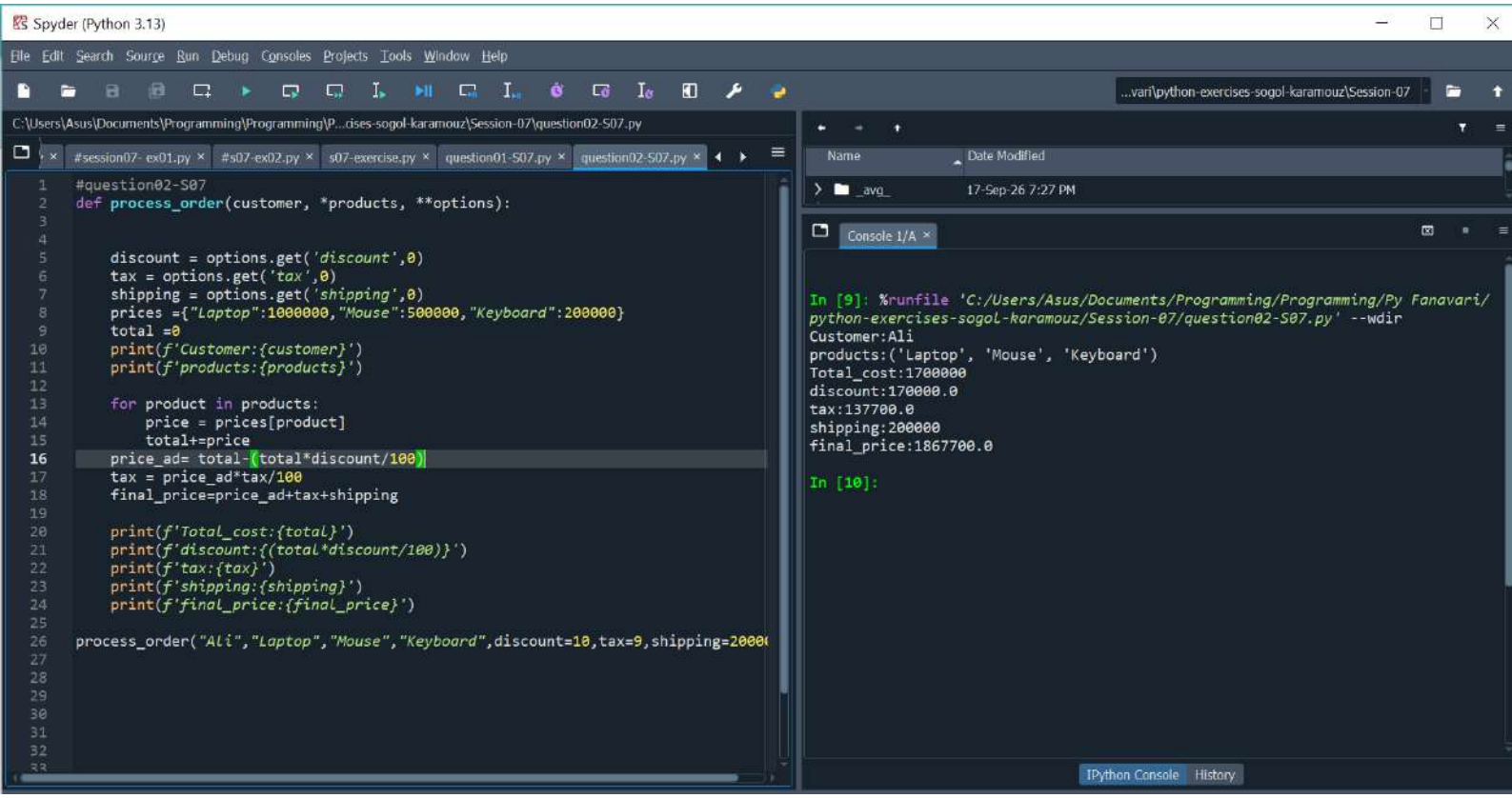
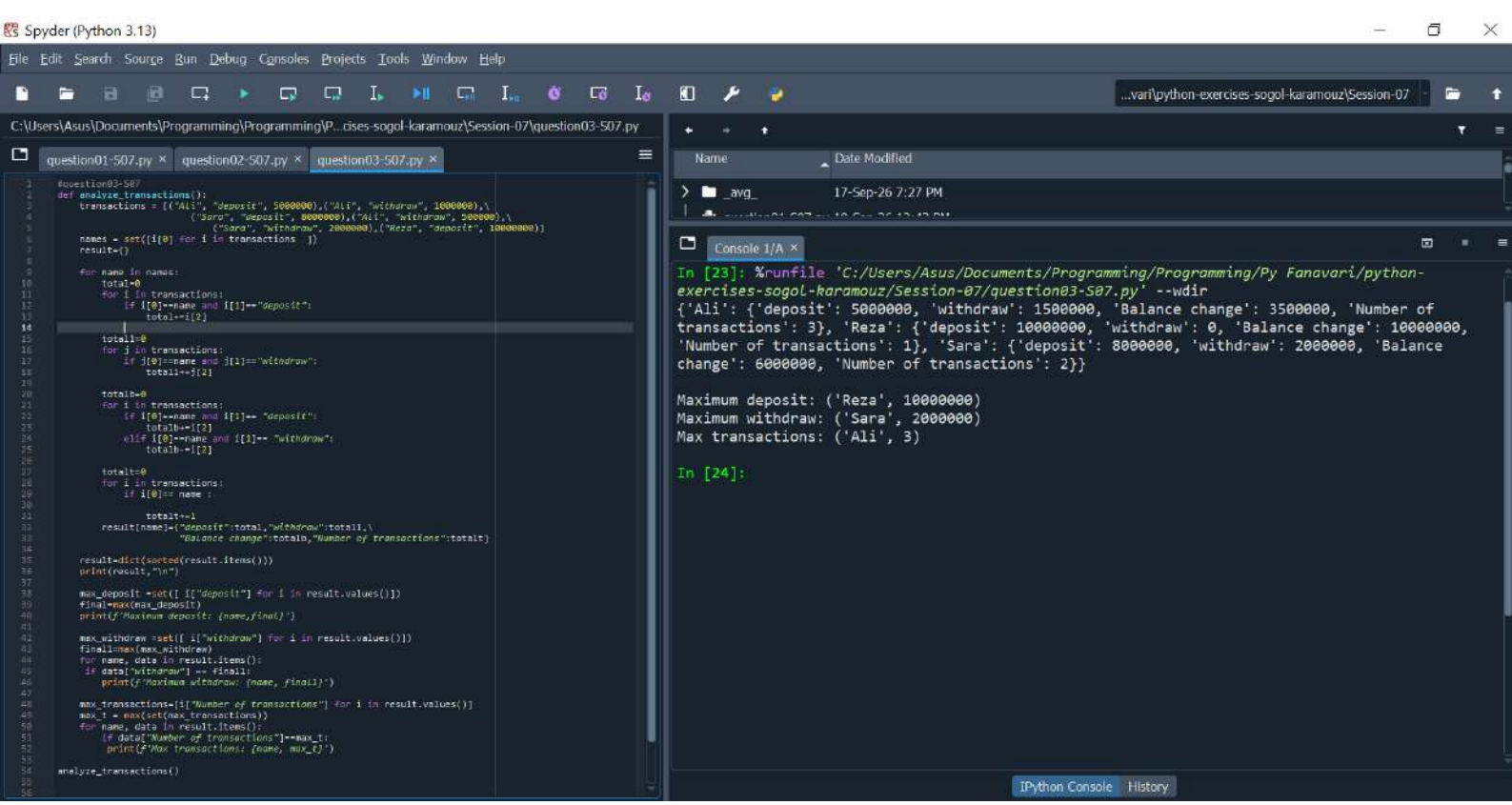
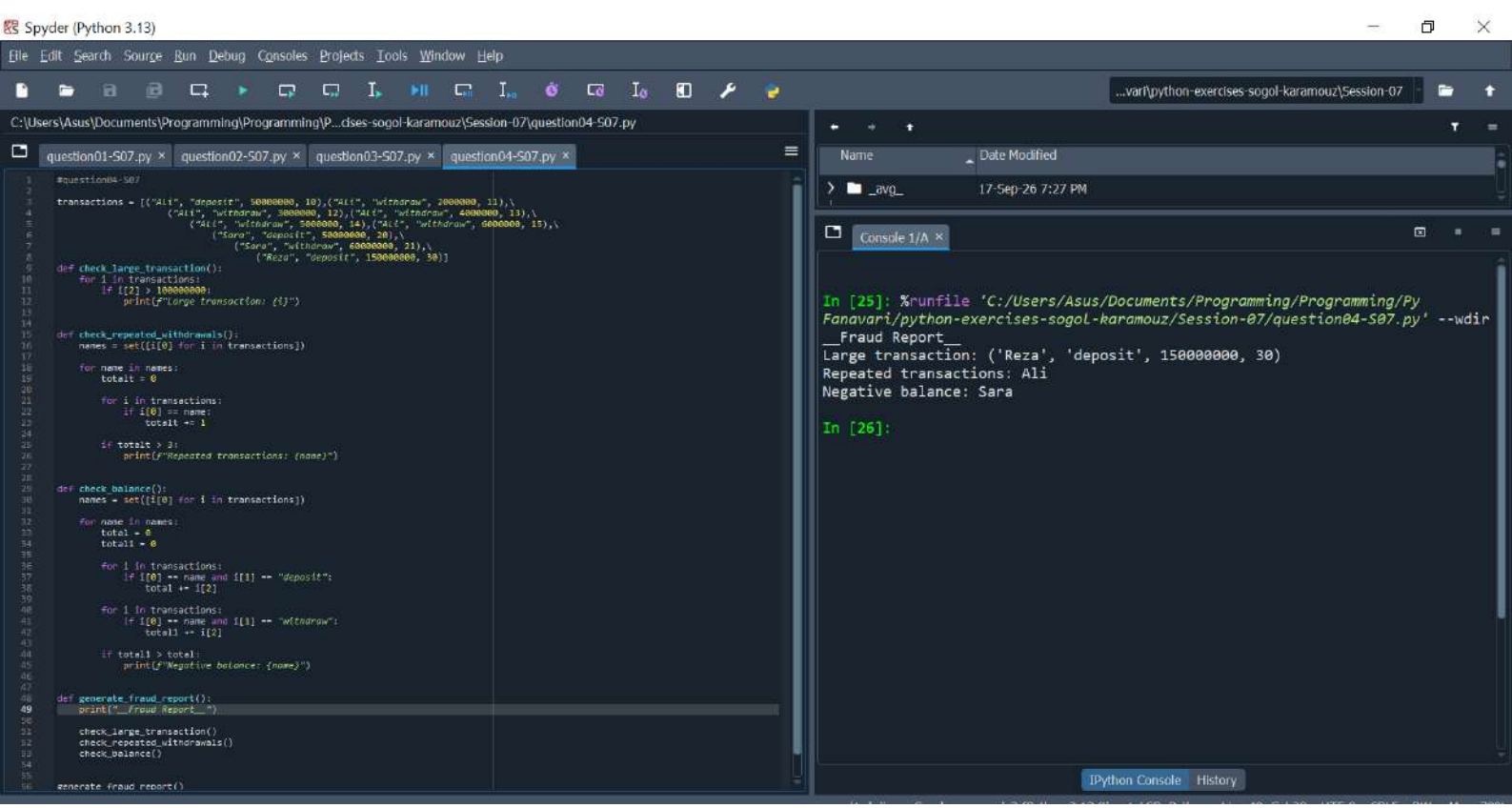
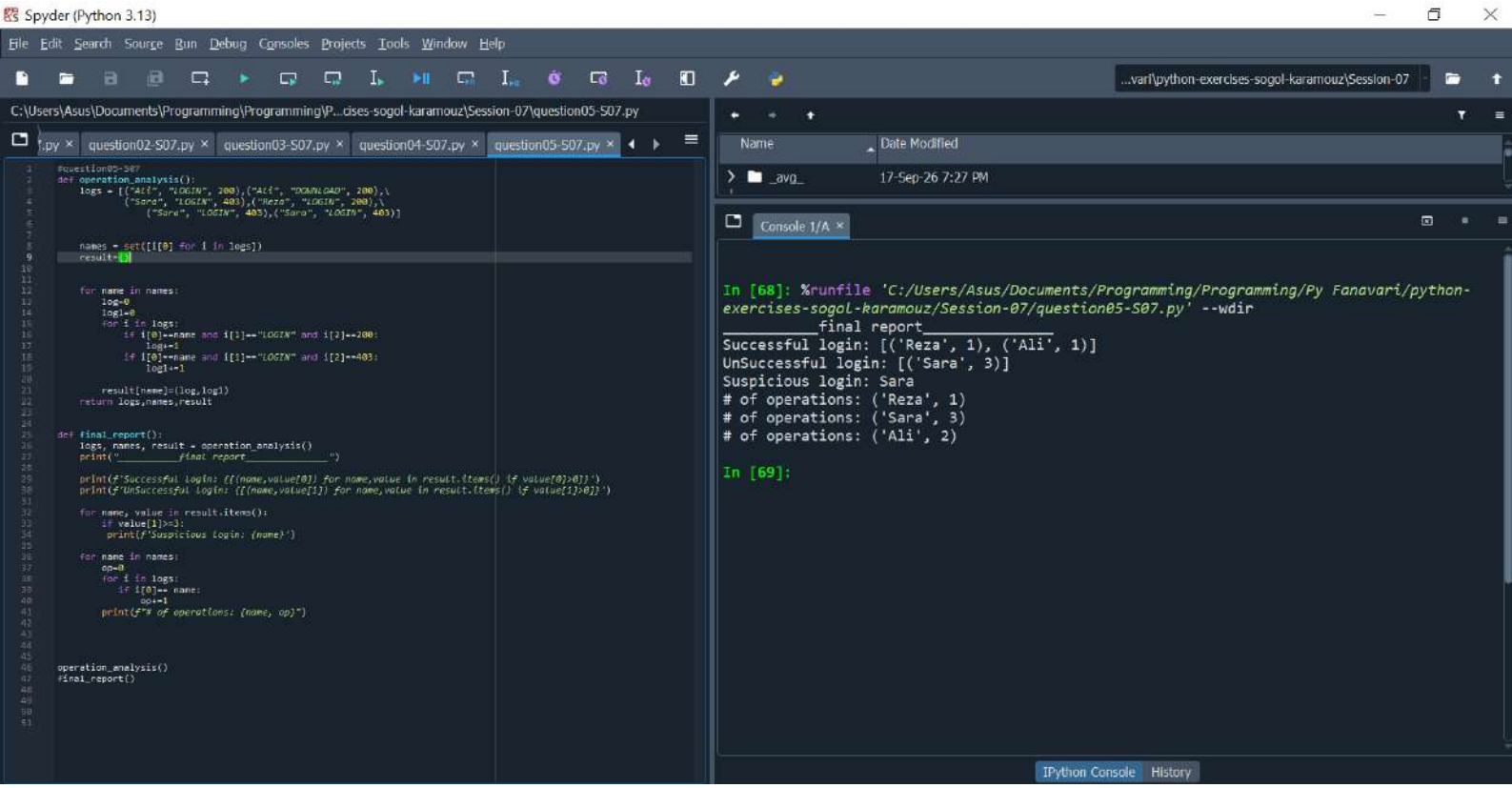

Spyder (Python 3.13)
File Edit Search Source Run Debug Consoles Projects Tools Window Help
...vari\python-exercises-sogol-karamouz\Session-07
C:\Users\Asus\Documents\Programming\Programming\P...cises-sogol-karamouz\Session-07\question01-S07.py
#session07- ex01.py
#s07-ex02.py
s07-exercise.py
question01-S07.py
#question01-S07
def analyse_text():
text = input('enter text: ')
words = len(text.split())
print(f'words: {words}')
letters =0
for j in text:
if j.isalpha():
letters+=1
print(f'letters: {letters}')
digits=0
for x in text:
if x.isdigit():
digits+=1
print(f'number of digits: {digits}')
word=text.split()
most_common_words = max(word, key=word.count)
print(f'most_common_words: {most_common_words}')
longest_word= max(word,key=len)
print(f'longest_word: {longest_word}')
shortest_word= min(word,key=len)
print(f'shortest_word: {shortest_word}')
letters = ''.join(word)
ml = max(letters.count(i) for i in letters)
most_common_letters = set(i for i in letters if letters.count(i)== ml)
print(f'most_common_letters: {most_common_letters}')
for i in word:
if i == i[::-1]:
print(f'pilandrome : {i}')
a=0
b=0
for i in text:
if i.isupper():
a +=1
elif i.islower():
b+=1
print(f'uperrcase:{a}')
print(f'lowercase:{b}')
analyse_text()
Name
Date Modified
avg
17-Sep-26 7:27 PM
Console 1/A
In [21]: %runfile 'C:/Users/Asus/Documents/Programming/Programming/Py Fanavari/python-exercises-sogol-karamouz/Session-07/question01-S07.py' --wdir
enter text: Python 3.2 is a level of Python
words: 7
letters: 22
number of digits: 2
most_common_words: Python
longest_word: Python
shortest_word: a
most_common_letters: {'o'}
pilandrome : a
pilandrome : level
uperrcase:2
lowercase:20
In [22]:
IPython Console
History


Spyder (Python 3.13)
File Edit Search Source Run Debug Consoles Projects Tools Window Help
...vari\python-exercises-sogol-karamouz\Session-07
C:\Users\Asus\Documents\Programming\Programming\P...cises-sogol-karamouz\Session-07\question02-S07.py
#session07- ex01.py
#s07-ex02.py
s07-exercise.py
question01-S07.py
question02-S07.py
#question02-S07
def process_order(customer, *products, **options):
discount = options.get('discount',0)
tax = options.get('tax',0)
shipping = options.get('shipping',0)
prices ={"Laptop":1000000,"Mouse":500000,"Keyboard":200000}
total =0
print(f'Customer:{customer}')
print(f'products:{products}')
for product in products:
price = prices[product]
total+=price
price_ad= total-(total*discount/100)
tax = price_ad*tax/100
final_price=price_ad+tax+shipping
print(f'Total_cost:{total}')
print(f'discount:{(total*discount/100)}')
print(f'tax:{tax}')
print(f'shipping:{shipping}')
print(f'final_price:{final_price}')
process_order("Ali","Laptop","Mouse","Keyboard",discount=10,tax=9,shipping=2000
Name
Date Modified
avg
17-Sep-26 7:27 PM
Console 1/A
In [9]: %runfile 'C:/Users/Asus/Documents/Programming/Programming/Py Fanavari/python-exercises-sogol-karamouz/Session-07/question02-S07.py' --wdir
Customer:Ali
products:('Laptop', 'Mouse', 'Keyboard')
Total_cost:1700000
discount:170000.0
tax:137700.0
shipping:200000
final_price:1867700.0
In [10]:
IPython Console
History


Spyder (Python 3.13)
File Edit Search Source Run Debug Consoles Projects Tools Window Help
...vari\python-exercises-sogol-karamouz\Session-07
C:\Users\Asus\Documents\Programming\Programming\P...cises-sogol-karamouz\Session-07\question03-S07.py
question01-S07.py
question02-S07.py
question03-S07.py
#question03-S07
def analyze_transactions():
transactions = [("Ali", "deposit", 5000000),("Ali", "withdraw", 1000000),\
("Sara", "deposit", 8000000),("Ali", "withdraw", 500000),\
("Sara", "withdraw", 2000000),("Reza", "deposit", 10000000)]
names = set([i[0] for i in transactions])
result={}
for name in names:
total=0
for I in transactions:
if I[0]==name and I[1]=="deposit":
total+=i[2]
totall=0
for j in transactions:
if j[0]==name and j[1]=="Withdraw":
totall+=j[2]
totalb=0
for i in transactions:
if i[0]==name and i[1]== "deposit":
totalb+=i[2]
elif i[0]==name and i[1]== "Withdraw":
totalb-=i[2]
totalt=0
for i in transactions:
if i[0]== name :
totalt+=1
result[name]={"deposit":total,"Withdraw":totall,\
"Balance change":totalb,"Number of transactions":totalt}
result=dict(sorted(result.items()))
print(result,"\n")
max_deposit =set([i["deposit"] for i in result.values()])
final=max(max_deposit)
print(f'Maximum deposit: {name,final}')
max_withdraw =set([i["withdraw"] for i in result.values()])
finall=max(max_withdraw)
for name, data in result.items():
if data["withdraw"] == finall:
print(f'Maximum withdraw: {name, finall}')
max_transactions=[i["Number of transactions"] for i in result.values()]
max_t = max(set(max_transactions))
for name, data in result.items():
if data["Number of transactions"]==max_t:
print(f'Max transactions: {name, max_t}')
analyze_transactions()
Name
Date Modified
avg
17-Sep-26 7:27 PM
Console 1/A
In [23]: %runfile 'C:/Users/Asus/Documents/Programming/Programming/Py Fanavari/python-exercises-sogol-karamouz/Session-07/question03-S07.py' --wdir
{'Ali': {'deposit': 5000000, 'withdraw': 1500000, 'Balance change': 3500000, 'Number of transactions': 3}, 'Reza': {'deposit': 10000000, 'withdraw': 0, 'Balance change': 10000000, 'Number of transactions': 1}, 'Sara': {'deposit': 8000000, 'withdraw': 2000000, 'Balance change': 6000000, 'Number of transactions': 2}}
Maximum deposit: ('Reza', 10000000)
Maximum withdraw: ('Sara', 2000000)
Max transactions: ('Ali', 3)
In [24]:
IPython Console
History


Spyder (Python 3.13)
File Edit Search Source Run Debug Consoles Projects Tools Window Help
...var\python-exercises-sogol-karamouz\Session-07
C:\Users\Asus\Documents\Programming\Programming\P...cises-sogol-karamouz\Session-07\question04-S07.py
question01-S07.py
question02-S07.py
question03-S07.py
question04-S07.py
#question04-S07
transactions = [("Ali", "deposit", 50000000, 10),("Ali", "withdraw", 2000000, 11),\
("Ali", "withdraw", 3000000, 12),("Ali", "withdraw", 4000000, 13),\
("Ali", "withdraw", 5000000, 14),("Ali", "withdraw", 6000000, 15),\
("Sara", "deposit", 50000000, 20),\
("Sara", "withdraw", 60000000, 21),\
("Reza", "deposit", 150000000, 30)]
def check_large_transaction():
for i in transactions:
if i[2] > 100000000:
print(f"Large transaction: {i}")
def check_repeated_withdrawals():
names = set([i[0] for i in transactions])
for name in names:
totalt = 0
for i in transactions:
if i[0] == name:
totalt += 1
if totalt > 3:
print(f"Repeated transactions: {name}")
def check_balance():
names = set([i[0] for i in transactions])
for name in names:
total = 0
total1 = 0
for i in transactions:
if i[0] == name and i[1] == "deposit":
total += i[2]
for i in transactions:
if i[0] == name and i[1] == "withdraw":
total1 += i[2]
if total1 > total:
print(f"Negative balance: {name}")
def generate_fraud_report():
print("__Fraud Report__")
check_large_transaction()
check_repeated_withdrawals()
check_balance()
generate_fraud_report()
Name
Date Modified
avg
17-Sep-26 7:27 PM
Console 1/A
In [25]: %runfile 'C:/Users/Asus/Documents/Programming/Programming/Py
Fanavari/python-exercises-sogol-karamouz/Session-07/question04-S07.py' --wdir
__Fraud Report__
Large transaction: ('Reza', 'deposit', 150000000, 30)
Repeated transactions: Ali
Negative balance: Sara
In [26]:
IPython Console
History


Spyder (Python 3.13)
File Edit Search Source Run Debug Consoles Projects Tools Window Help
...vari\python-exercises-sogol-karamouz\Session-07
C:\Users\Asus\Documents\Programming\Programming\P...cises-sogol-karamouz\Session-07\question05-S07.py
question02-S07.py
question03-S07.py
question04-S07.py
question05-S07.py
#question05-S07
def operation_analysis():
logs = [("Ali", "LOGIN", 200),("Ali", "DOWNLOAD", 200),\
("Sara", "LOGIN", 403),("Reza", "LOGIN", 200),\
("Sara", "LOGIN", 403),("Sara", "LOGIN", 403)]
names = set([i[0] for i in logs])
result={}
for name in names:
log=0
log1=0
for i in logs:
if i[0]==name and i[1]=="LOGIN" and i[2]==200:
log+=1
if i[0]==name and i[1]=="LOGIN" and i[2]==403:
log1+=1
result[name]=(log,log1)
return logs,names,result
def final_report():
logs, names, result = operation_analysis()
print("_____final report_____")
print(f'Successful login: {[(name,value[0]) for name,value in result.items() if value[0]>0]}')
print(f'UnSuccessful login: {[(name,value[1]) for name,value in result.items() if value[1]>0]}')
for name, value in result.items():
if value[1]>=3:
print(f'Suspicious login: {name}')
for name in names:
op=0
for i in logs:
if i[0]== name:
op+=1
print(f"# of operations: {name, op}")
operation_analysis()
final_report()
Name
Date Modified
avg
17-Sep-26 7:27 PM
Console 1/A
In [68]: %runfile 'C:/Users/Asus/Documents/Programming/Programming/Py Fanavari/python-exercises-sogol-karamouz/Session-07/question05-S07.py' --wdir
_____final report_____
Successful login: [('Reza', 1), ('Ali', 1)]
UnSuccessful login: [('Sara', 3)]
Suspicious login: Sara
of operations: ('Reza', 1)
of operations: ('Sara', 3)
of operations: ('Ali', 2)
In [69]:
IPython Console
History